BYoung Physical Therapy

Patient: John Jiles • **Date:** November 09, 2026
Surgical Side: Left • **Surgery:** 04/21/2026 (6.6 mo post-op) • **Triple Hop:** Surg 22'0" / NS 22'7" / LSI 97.4%

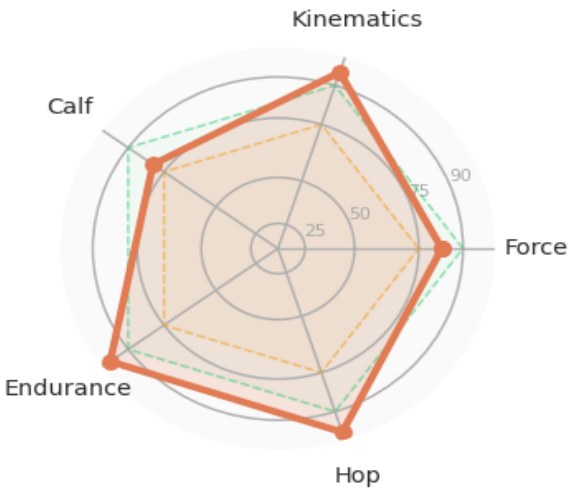
Return to Sport Assessment
Achilles Repair

38

RTS Score
Not Ready

5 scored red flags · 6.6 mo post-op

Return to Sport Profile



▲ STRENGTHS

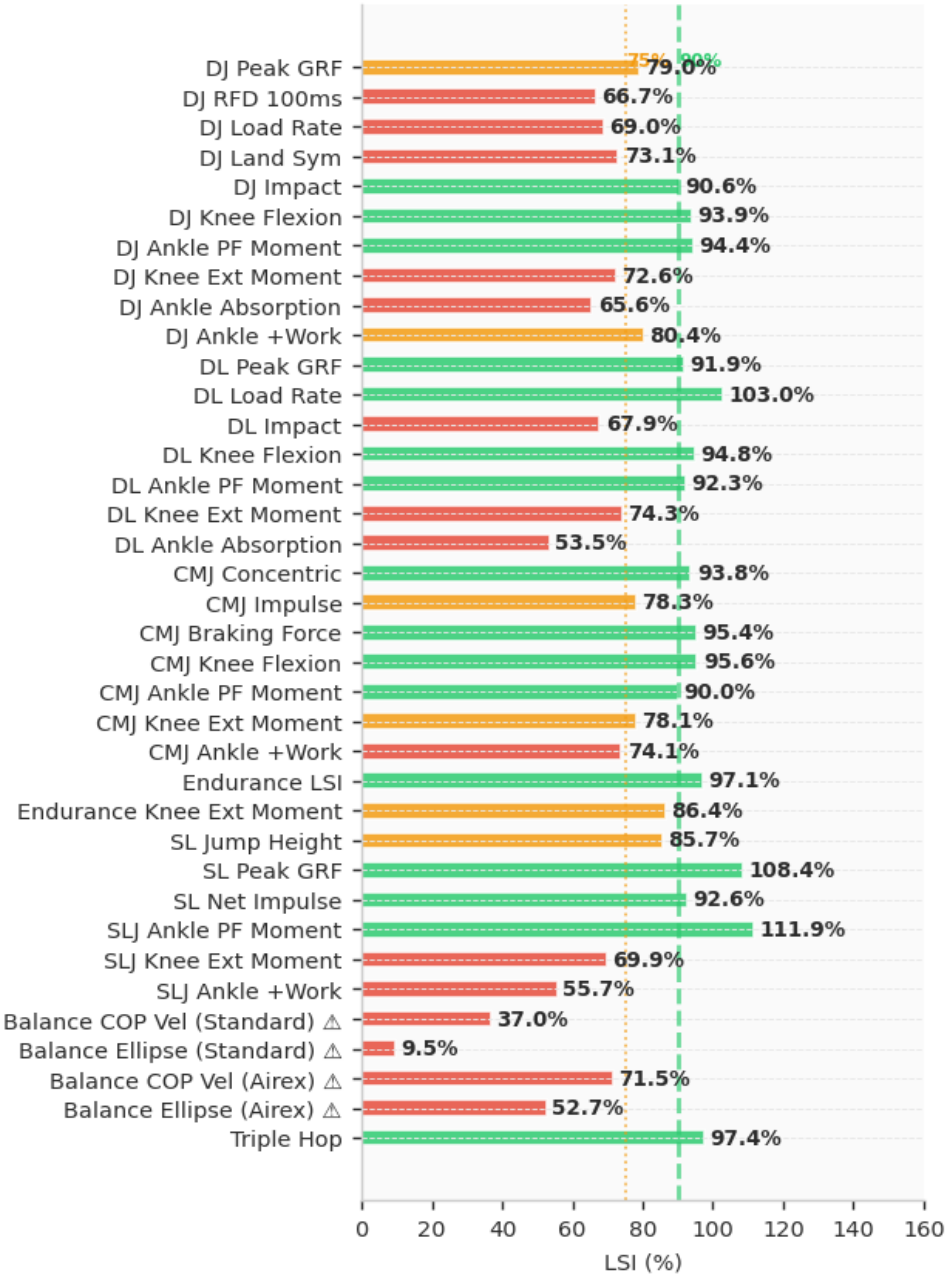
- Ankle PF moment sym. (100/100)
- DL Load rate sym. (100/100)
- Fatigue drift (100/100)
- Endurance symmetry (91/100)

▼ DETRACTORS

- Balance symmetry (30/100)
- Ankle push-off work (SLJ) (40/100)
- Landing symmetry (48/100)
- Rate of force dev. (48/100)
- DJ Load rate sym. (49/100)

Performance 65/100 → ×0.71 red-flag penalty (5 scored) → ×0.82 time modifier → = 38/100

Limb Symmetry Index — All Tests



CLINICAL FLAGS

- SLJ Ankle Dorsiflexion: Δ -8.8° — exceeds 6° threshold
- DL Ankle Absorption: LSI 53.5% — significant asymmetry
- SLJ Ankle +Work: LSI 55.7% — significant asymmetry
- DJ Ankle Absorption: LSI 65.6% — significant asymmetry
- DJ RFD 100ms: LSI 66.7% — significant asymmetry
- DL Impact: LSI 67.9% — significant asymmetry

• +6 more — see individual test pages.

Achilles profile — rate-of-loading, reactive and calf-capacity metrics are weighted above peak force. Knee valgus / tibial rotation are shown for context, not scored.
Balance (standard) excluded from the score — COP data check failed (foot shift / touchdown); re-collect.
Balance (airex) excluded from the score — COP data check failed (foot shift / touchdown); re-collect.
No single-leg heel-raise result entered — calf capacity is the primary Achilles criterion and is not represented in this score.
Body weight from force plates is 944 N (212 lb) but the Visual3D session mass gives 845 N (190 lb). Plate value used; check the session weight in Visual3D (it also scales joint moments).

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100 | Δ = Surgical – Non-Surgical (absolute difference; used for small-angle kinematics)

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DROP JUMP

ABOUT THIS TEST

The Drop Jump is a reactive bilateral task: the athlete drops from a box and immediately jumps as high and as quickly as possible. It loads the calf–Achilles unit through a fast stretch-shortening cycle — the same demand as sprinting and cutting. After Achilles repair the common compensations are a longer ground contact, a slower and lower initial force rise on the surgical limb (reduced RFD / loading rate), a heel-biased landing, and shifting work to the knee and hip. The Reactive Strength Index (jump height ÷ contact time) summarises how well the athlete uses elastic energy; contact time itself shows whether the jump was truly reactive (< 0.25 s) or a slow, guarded rebound.

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1311.8	1659.9	79.0%
Impulse (N·s)	464.1	552.4	84.0%
RFD 100ms (N/s)	19666.6	29483.9	66.7%
Loading Rate 50ms (N/s)	17307.7	25071.5	69.0%
Landing Sym 200ms (%) *(bilateral)*	—	—	73.1%
Impact Transient (N)	815.7	900.2	90.6%
Time-to-Peak (ms)	167.4	70.7	—
Jump Height (in)	24.0	—	—
Contact Time (s)	0.481	—	—
RSI (m/s)	1.27	—	—
Peak Knee Flexion (°)	85.9	91.4	93.9%
Peak Knee Valgus (°)	8.2	3.0	Δ +5.2°
Peak Tibial IR (°)	-6.4	-0.4	Δ -6.0°
Tibial Rot Excursion (°)	6.4	2.7	Δ +3.7°
Peak Hip Flexion (°)	79.6	82.9	Δ -3.2°
Peak Hip Adduction (°)	-2.8	-0.4	Δ -2.4°
Peak Hip IR (°)	11.7	4.4	Δ +7.3°
Peak Ankle Dorsiflexion (°)	26.4	29.5	Δ -3.1°
Peak Plantarflexor Moment (Nm)	174.8	185.2	94.4%
Peak Knee Ext Moment (Nm)	143.6	197.8	72.6%
AP COP Excursion (mm)	116.9	134.0	—
Trials analysed	3	—	—
Ankle Absorption (J)	50.1	76.4	65.6%
Knee Absorption (J)	62.3	119.9	51.9%
Hip Absorption (J)	98.4	118.6	83.0%
Ankle share of absorption (%)	23.2	24.1	—
Ankle +Work (J)	78.8	98.0	80.4%

Knee +Work (J)	82.5	98.3	84.0%
Hip +Work (J)	159.5	163.6	97.5%
Ankle share of +work (%)	24.4	27.1	—

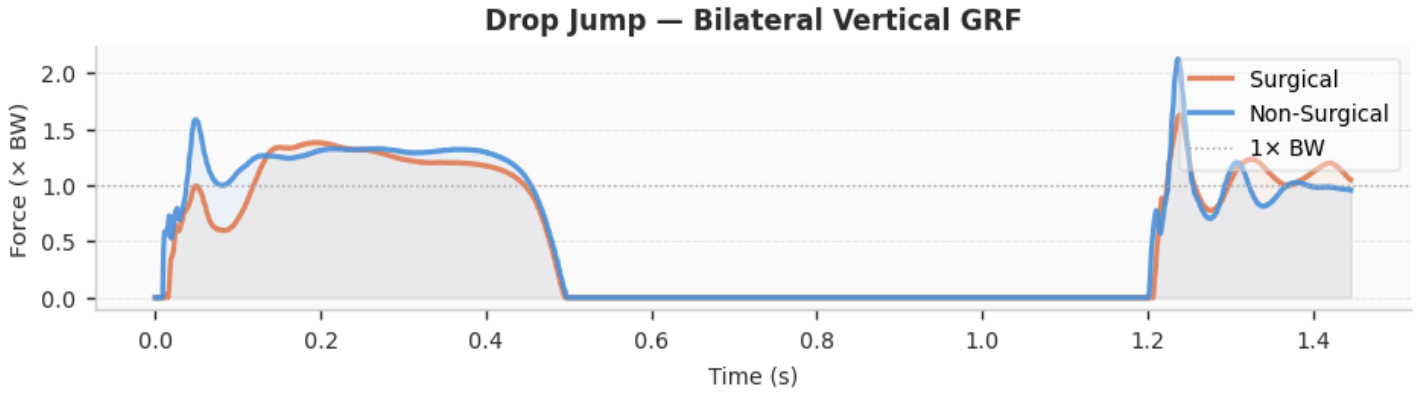
CLINICAL INTERPRETATION

RSI 1.27 m/s with 0.48 s contact (⚠ below RTS target ≥ 1.30). Long contact / low reactive strength — the surgical calf–Achilles unit is not yet tolerating fast stretch-shortening loads. Progress pogo → short-contact drop jumps.

Contact time 0.48 s — not a reactive (fast stretch-shortening) contact; target < 0.25 s.

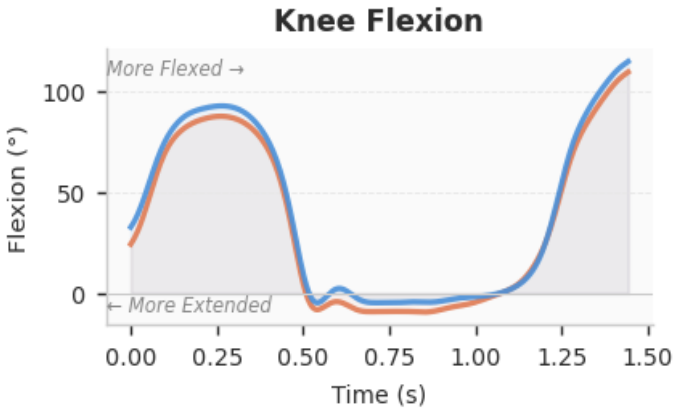
Landing symmetry 73% LSI: Surgical limb loading 27% less than non-surgical — surgical guarding detected.

Targets: RSI ≥ 1.30 m/s · Contact < 0.25 s · RFD & load rate LSI ≥ 85–90% · Landing LSI 90–110%.

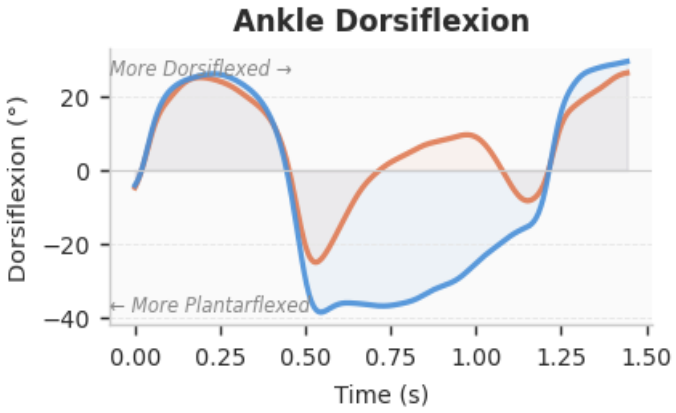


Bilateral GRF (normalised to BW). A slower, lower initial rise on the surgical side = reduced fast loading through the calf–Achilles unit.

Landing symmetry 73% at 200ms — surgical limb offloading detected.

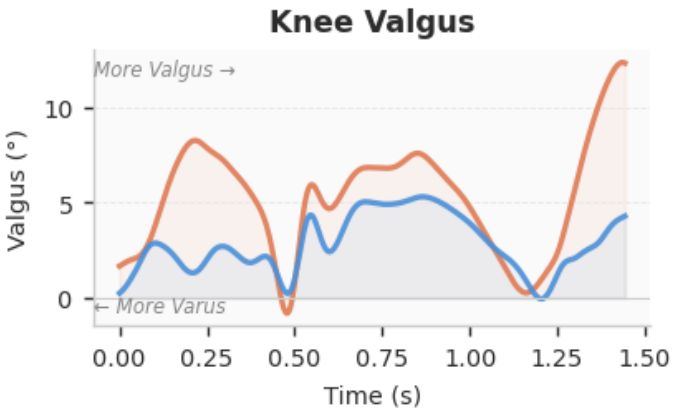


Knee flexion. Deeper knee bend on landing can be used to offload the ankle.

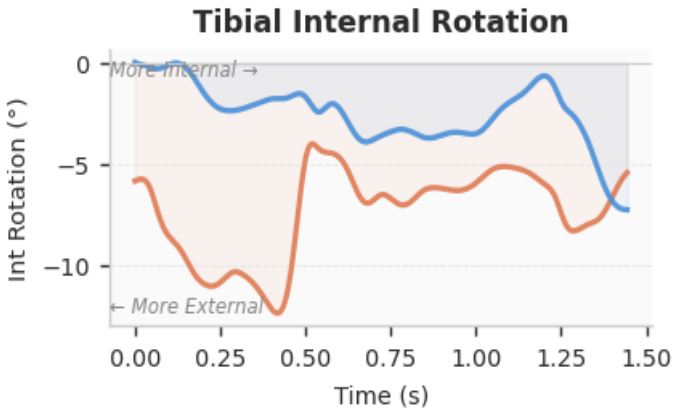


Ankle dorsiflexion. Reduced surgical dorsiflexion under load = calf guarding; excess = tendon elongation / reduced stiffness.

***Left panel:** Surgical limb knee flexion depth near-symmetric (LSI 94%) — approaching return-to-sport target; minor deficit remains.*



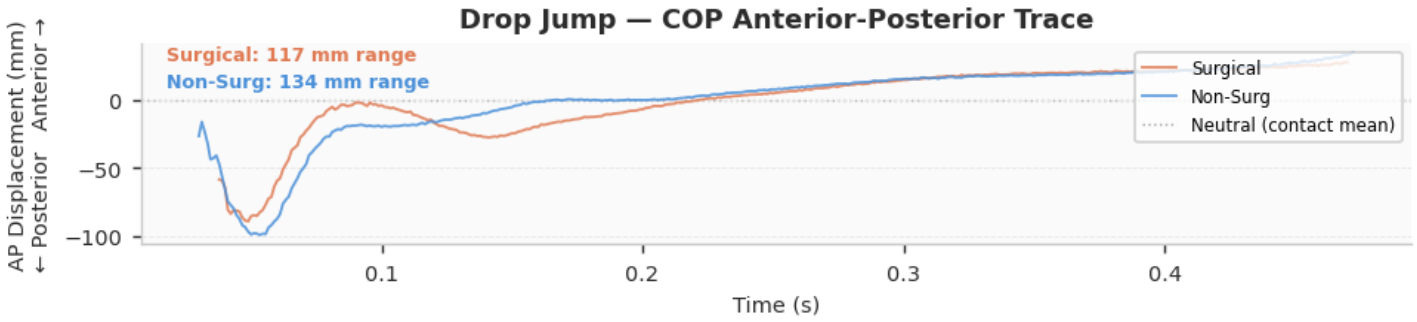
Knee valgus (context only for Achilles — not scored).



Tibial / foot rotation. A consistent surgical-side external-rotation offset can reflect toe-out (avoiding push-off through the forefoot) or a static calibration offset — confirm with video.

Left panel: *Surgical knee valgus: 8.2° vs non-surgical 3.0° (Δ +5.2°) — mild side-to-side asymmetry (context only — not scored for this report type).*

Right panel: *Surgical tibial IR: -6.4° vs non-surgical -0.4° (Δ -6.0°) — mild side-to-side asymmetry (context only — not scored for this report type).*



AP COP during the first contact (zero-centred). Posterior / heel bias on the surgical side = reduced forefoot loading through the Achilles.

AP COP excursion: surgical 117 mm vs non-surgical 134 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb bilateral landing symmetry demonstrates a significant deficit (LSI 73%) — targeted strengthening and re-testing indicated.

DROP LANDING

ABOUT THIS TEST

The Drop Landing isolates load acceptance: the athlete drops from a box and sticks the landing without a re-jump. After Achilles repair the surgical limb often lands heel-first and accepts load more slowly, which shows up as a lower loading rate and impact transient rather than a lower peak force. Symmetric loading rate means the calf is eccentrically controlling the forefoot-to-heel transition on both sides.

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1339.1	1456.6	91.9%
Impulse (N·s)	621.7	640.8	97.0%
Loading Rate 50ms (N/s)	23242.3	22571.8	103.0%
Impact Transient (N)	708.4	1043.4	67.9%
Peak Knee Flexion (°)	111.1	117.3	94.8%
Peak Knee Valgus (°)	10.8	4.2	Δ +6.6°
Peak Tibial IR (°)	-4.6	1.0	Δ -5.5°
Tibial Rot Excursion (°)	5.4	4.3	Δ +1.1°
Peak Hip Flexion (°)	106.1	105.6	Δ +0.5°
Peak Hip Adduction (°)	-5.8	-7.8	Δ +2.0°
Peak Hip IR (°)	16.7	8.9	Δ +7.8°
Peak Ankle Dorsiflexion (°)	21.2	27.8	Δ -6.6°
Peak Plantarflexor Moment (Nm)	135.9	147.3	92.3%
Peak Knee Ext Moment (Nm)	102.0	137.2	74.3%
AP COP Excursion (mm)	73.3	106.4	—
Trials analysed	3	—	—
Ankle Absorption (J)	49.6	92.8	53.5%
Knee Absorption (J)	81.1	145.4	55.8%
Hip Absorption (J)	209.3	162.1	129.1%
Ankle share of absorption (%)	14.7	23.3	—

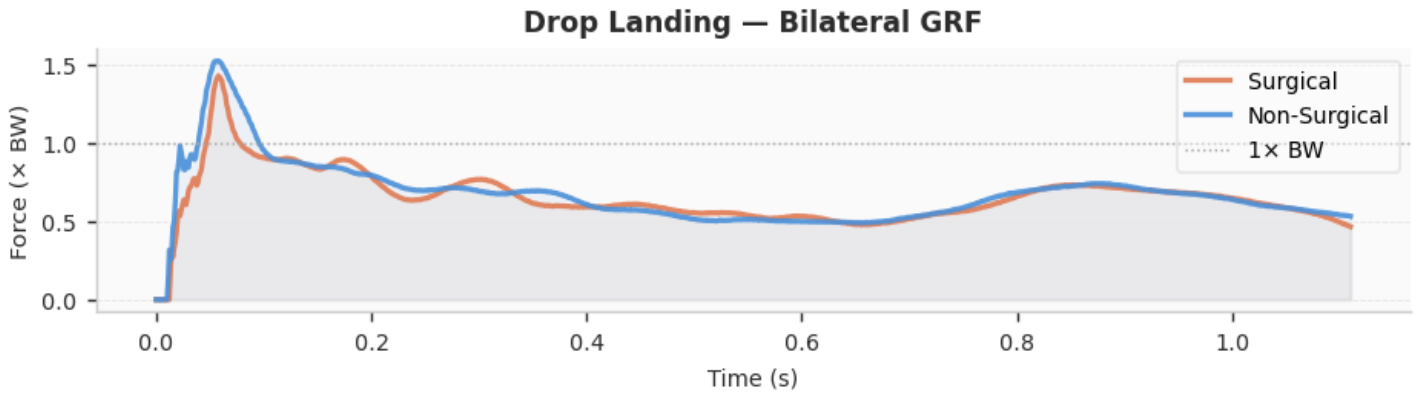
CLINICAL INTERPRETATION

Absorptive landing — assesses how quickly and symmetrically the surgical limb accepts load.

Load rate LSI 103%: Symmetric loading strategy.

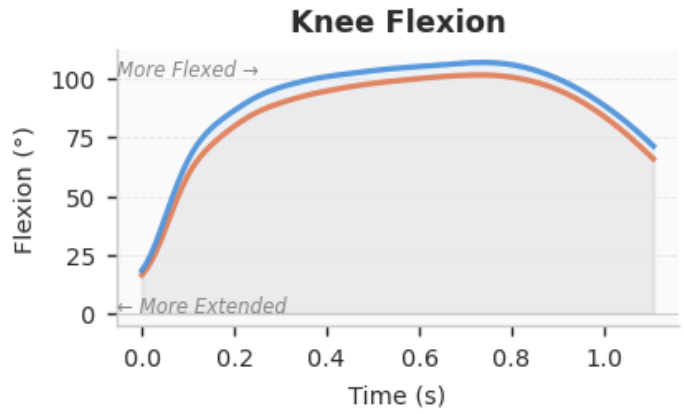
Impact transient LSI 68% — surgical limb initial impact 32% lower.

Targets: Load rate LSI ≥ 85–90% · Impact transient LSI 90–110% · Peak GRF LSI ≥ 90%.

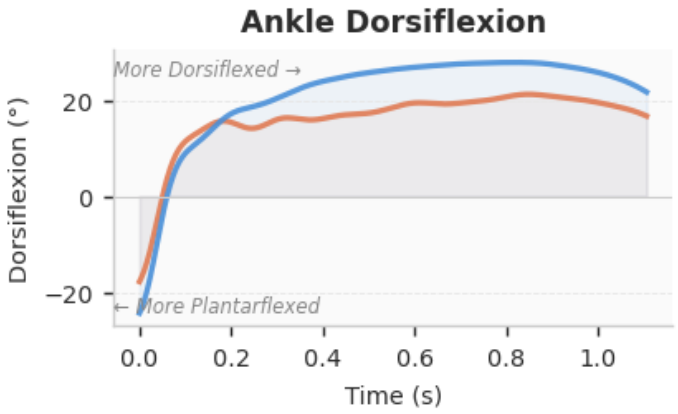


Bilateral GRF. Lower, later initial peak on the surgical side = slower load acceptance.

Peak GRF symmetry 92% — approaching return-to-sport target; minor deficit remains.

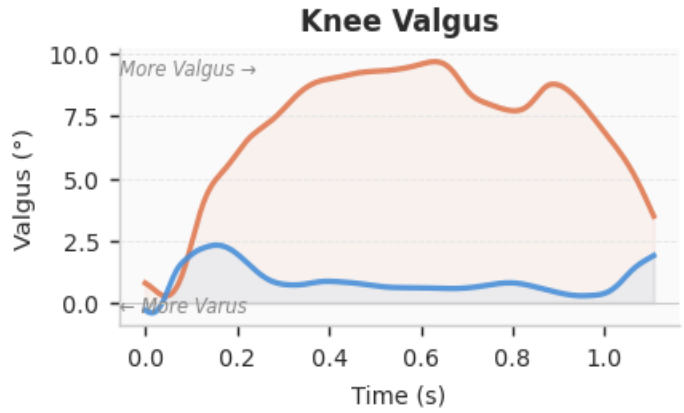


Knee flexion at landing. Extra knee bend can substitute for ankle absorption.

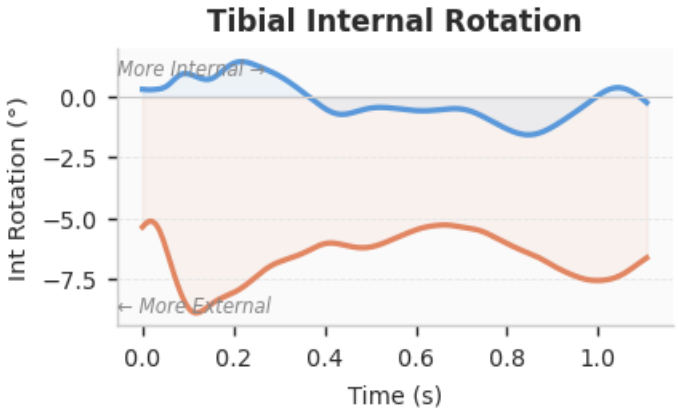


Ankle dorsiflexion at landing — eccentric calf control.

***Left panel:** Surgical limb landing knee flexion depth near-symmetric (LSI 95%) — approaching return-to-sport target; minor deficit remains.*



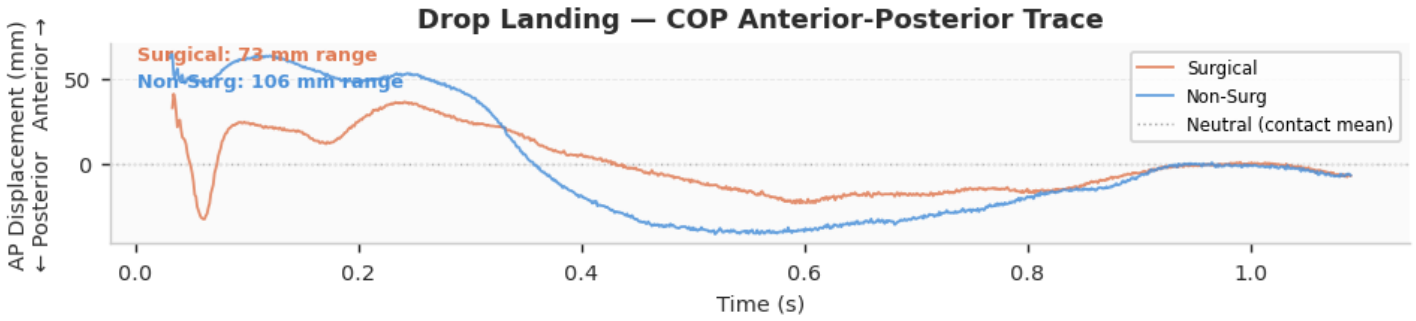
Knee valgus at landing (context only for Achilles — not scored).



Tibial / foot rotation. Surgical-side external rotation offset = possible toe-out strategy.

***Left panel:** Surgical knee valgus: 10.8° vs non-surgical 4.2° (Δ +6.6°) — side-to-side difference exceeds 6° (context only — not scored for this report type).*

***Right panel:** Surgical tibial IR: -4.6° vs non-surgical 1.0° (Δ -5.5°) — mild side-to-side asymmetry (context only — not scored for this report type).*



AP COP during landing contact (zero-centred). Posterior bias on surgical side = heel-dominant / reduced forefoot loading.

AP COP excursion: surgical 73 mm vs non-surgical 106 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb peak landing force absorption near-symmetric (LSI 92%) — approaching return-to-sport target; minor deficit remains.

MAX VERTICAL JUMP (CMJ)

ABOUT THIS TEST

The Countermovement Jump is a bilateral power test with a relatively long, slow stretch-shortening cycle. Because the hip and knee can contribute most of the work, CMJ symmetry often returns before the calf has recovered — a symmetric CMJ does not clear an Achilles. Propulsive impulse and peak force LSI show whether the surgical limb is contributing to push-off.

Metric	Surgical	Non-Surg	LSI / Δ
Peak Propulsive Force (N)	1216.9	1296.9	93.8%
Net Propulsive Impulse (N·s)	148.8	190.0	78.3%
Peak Braking Force (N)	1217.9	1276.0	95.4%
Jump Height (in)	25.0	—	—
Flight Time (s)	0.720	—	—
Unweighting Impulse Surg/NS	57.4	58.8	—
Peak Knee Flexion (°)	84.1	88.0	95.6%
Peak Knee Valgus (°)	10.1	2.1	Δ +8.0°
Peak Tibial IR (°)	-4.9	2.7	Δ -7.6°
Tibial Rot Excursion (°)	7.4	5.9	Δ +1.5°
Peak Hip Flexion (°)	96.7	98.0	Δ -1.4°
Peak Hip Adduction (°)	-3.1	2.2	Δ -5.3°
Peak Hip IR (°)	11.4	1.3	Δ +10.1°
Peak Ankle Dorsiflexion (°)	27.2	28.3	Δ -1.1°
Peak Plantarflexor Moment (Nm)	163.2	181.4	90.0%
Peak Knee Ext Moment (Nm)	122.7	157.0	78.1%
AP COP Excursion (mm)	138.5	30.4	—
Trials analysed	2	—	—
Braking Impulse (N·s)	49.1	73.9	66.5%
Ankle +Work (J)	85.0	114.8	74.1%
Knee +Work (J)	69.2	98.4	70.3%
Hip +Work (J)	266.4	234.3	113.7%
Ankle share of +work (%)	20.2	25.7	—

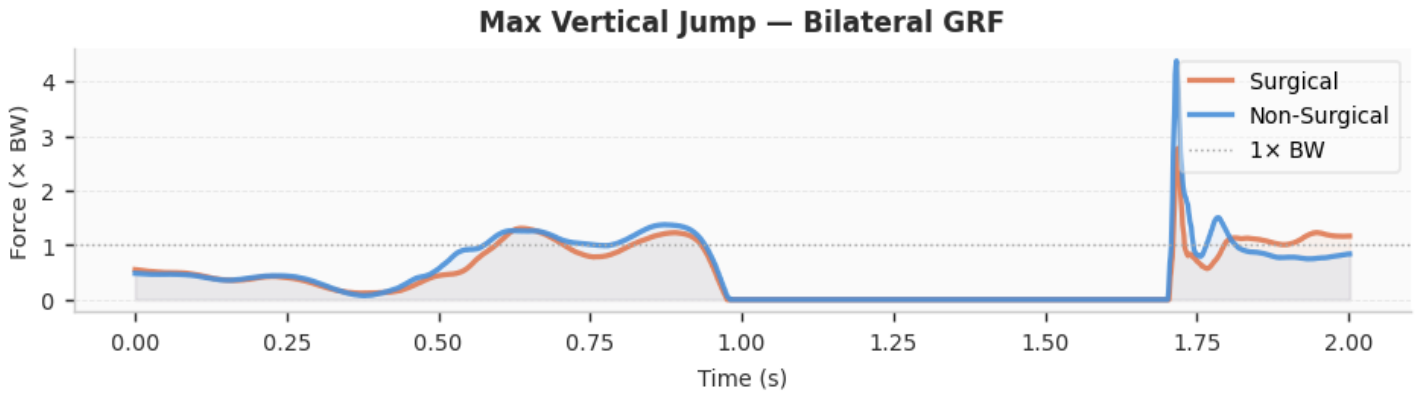
CLINICAL INTERPRETATION

Bilateral CMJ — slow stretch-shortening; usually recovers before fast-SSC tasks after Achilles repair.

Braking force LSI 95%: Symmetric bilateral output.

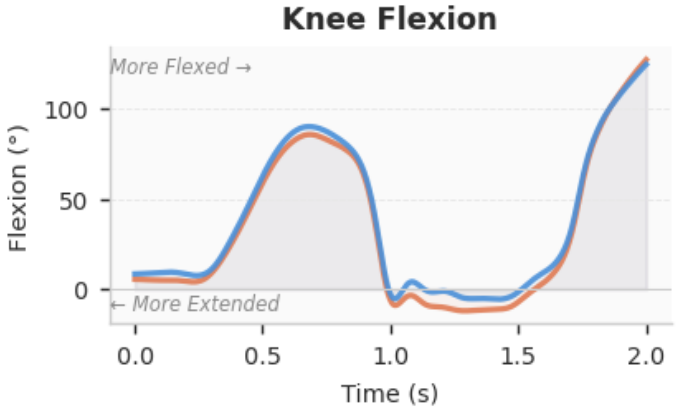
Propulsion LSI 78%. Targets: ≥ 90% propulsion, ≥ 90% peak force.

Ankle +work LSI 74% vs hip 114%: The surgical ankle contributes less push-off work while the hip makes up the difference — the calf is being bypassed in the propulsive phase.

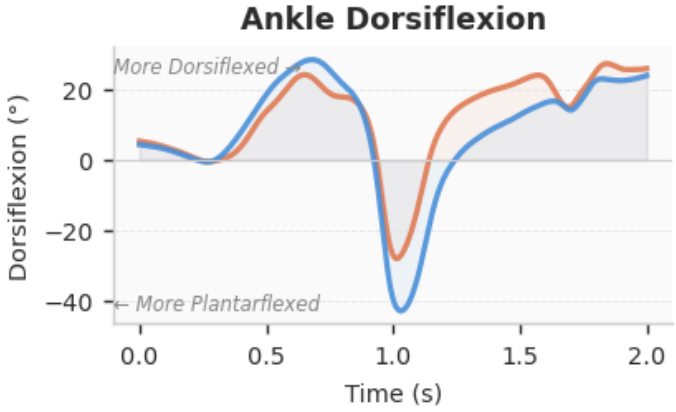


Full CMJ: unweighting → eccentric → propulsion → flight. Propulsive peaks should be symmetric.

Surgical limb propulsive impulse shows a moderate asymmetry (LSI 78%) — continued loading progression recommended before clearance.

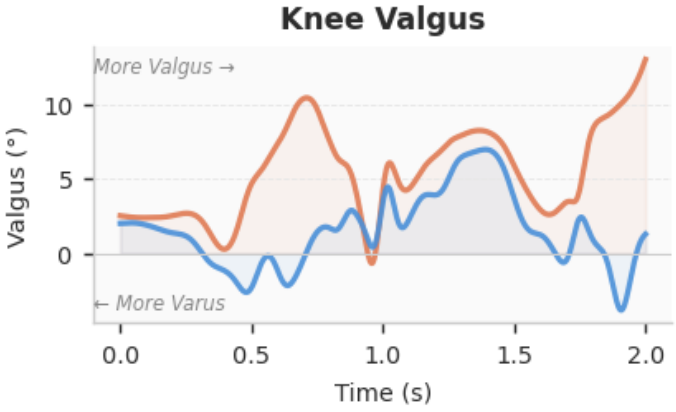


Knee flexion depth. Asymmetric countermovement depth = compensatory loading strategy.

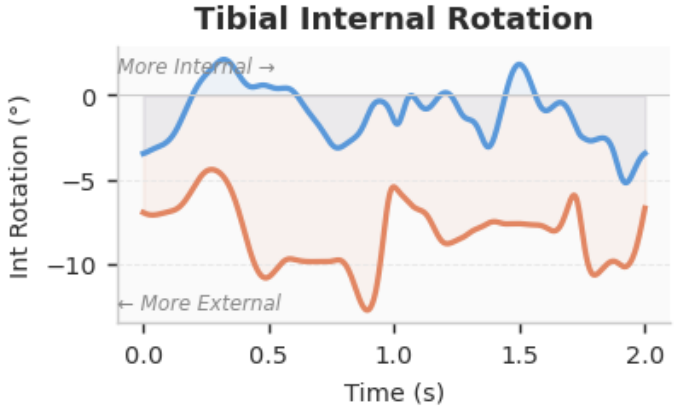


Ankle dorsiflexion — the final push-off through plantarflexion drives jump height.

***Left panel:** Surgical limb countermovement knee flexion depth is symmetric with the non-surgical side (LSI 96%) — meets return-to-sport threshold.*



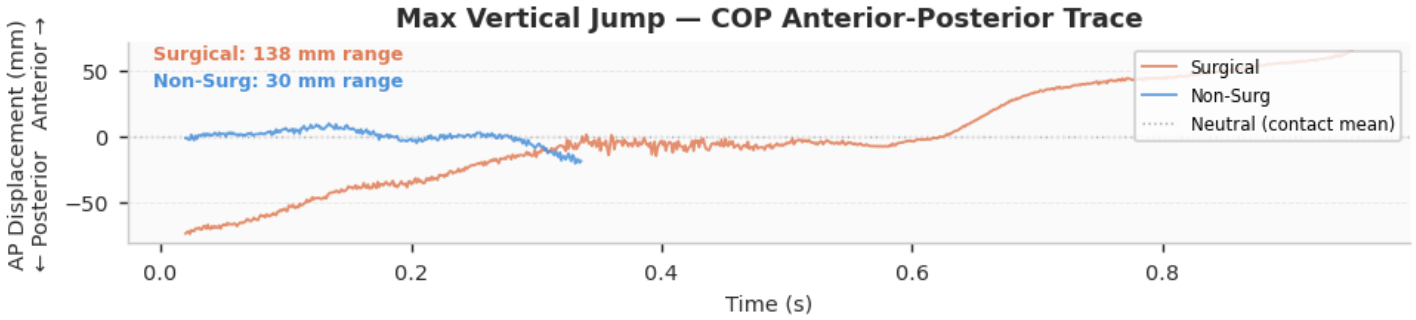
Knee valgus (context only for Achilles — not scored).



Tibial / foot rotation. Surgical-side external rotation offset = possible toe-out strategy.

***Left panel:** Surgical knee valgus: 10.1° vs non-surgical 2.1° (Δ +8.0°) — side-to-side difference exceeds 6° (context only — not scored for this report type).*

***Right panel:** Surgical tibial IR: -4.9° vs non-surgical 2.7° (Δ -7.6°) — mild side-to-side asymmetry (context only — not scored for this report type).*



AP COP during the propulsive contact (zero-centred). Less anterior travel on the surgical side = reduced push-off through the forefoot.

AP COP excursion: surgical 138 mm vs non-surgical 30 mm — more fore-aft travel on the surgical side — less controlled foot loading.

Surgical limb peak propulsive force is symmetric with the non-surgical side (LSI 95%) — meets return-to-sport threshold.

ENDURANCE SQUAT (30s)

ABOUT THIS TEST

The 30-Second Endurance Squat assesses limb symmetry under sustained bilateral loading. It is knee- and hip-dominant, so for an Achilles repair it mainly checks for whole-limb offloading under fatigue; calf endurance is assessed with the single-leg heel raise to failure.

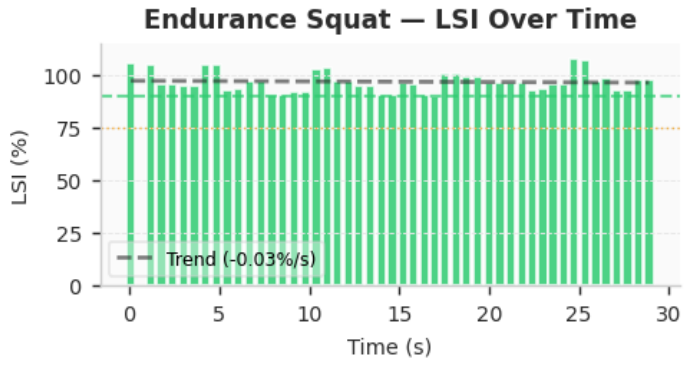
Metric	Surgical	Non-Surg	LSI / Δ
Mean Peak Force LSI (%) *(bilateral)*	—	—	97.1%
LSI First 10s (%) *(bilateral)*	—	—	97.0%
LSI Last 10s (%) *(bilateral)*	—	—	97.4%
Fatigue Drift (Δ%)	0.4	—	—
Total Reps Detected	48.0	—	—
Mean Peak Force Surg (N)	903.3	—	—
Mean Peak Force NS (N)	931.9	—	—
AP COP Excursion — Surg / NS (mm)	218.2	225.8	—
Peak Knee Ext Moment (Nm)	123.6	143.1	86.4%

CLINICAL INTERPRETATION

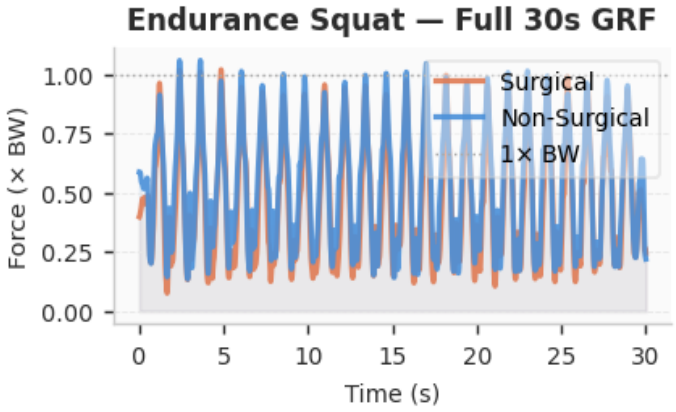
30s bilateral squat — knee-dominant endurance; does not directly load the calf.

Fatigue drift 0.4%: ✓ Symmetry maintained under fatigue.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5%. Pair with the heel-raise test for calf endurance.

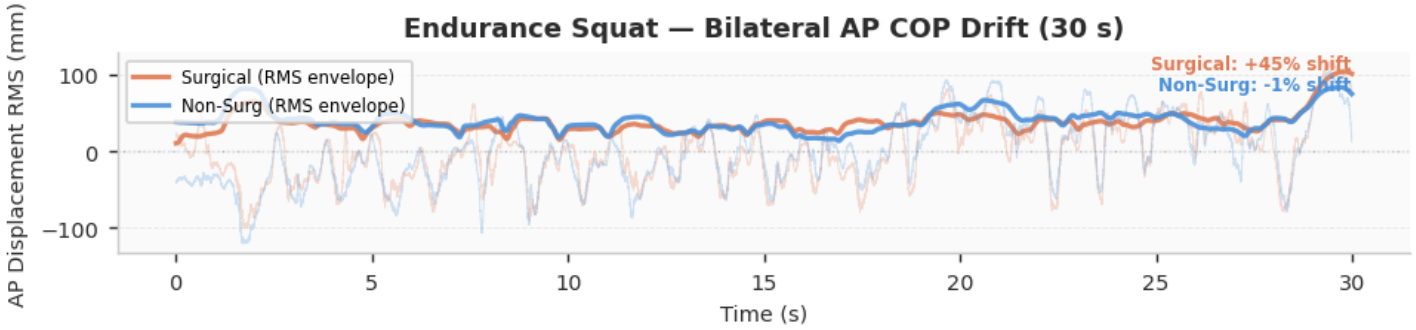


Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

***Left panel:** Surgical limb mean 30s force symmetry is symmetric with the non-surgical side (LSI 97%) — meets return-to-sport threshold.*



Bilateral AP COP drift across 30s (zero-centred, 1s RMS envelope). Surgical-side amplitude shrinking or trending posterior = offloading under fatigue. Diverging traces = progressive asymmetry with sustained effort.

Fatigue drift: 0.4% LSI change across 30s — symmetry maintained under sustained effort.

Surgical limb mean force symmetry over 30 s is symmetric with the non-surgical side (LSI 97%) — meets return-to-sport threshold.

SINGLE-LEG VERTICAL JUMP

ABOUT THIS TEST

The Single-Leg Vertical Jump isolates each limb so the uninvolved side cannot compensate. Final push-off depends on the plantarflexors, so after Achilles repair single-leg height is one of the more sensitive power measures. Compare jump height with net impulse: if impulse is near-symmetric but height is not, the athlete is producing force slowly over a longer contact rather than quickly through the calf.

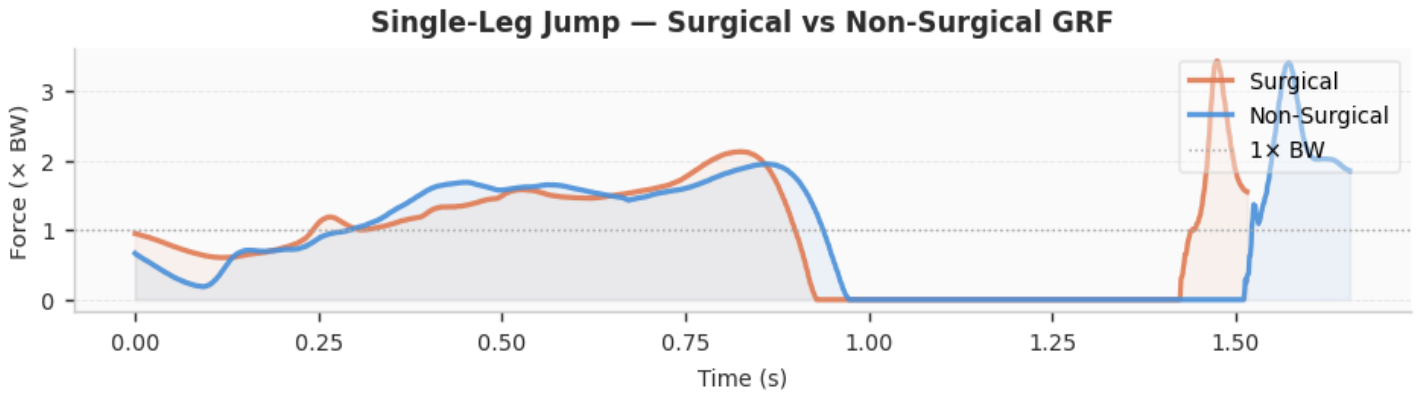
Metric	Surgical	Non-Surg	LSI / Δ
Jump Height (in)	12.0	14.0	85.7%
Peak GRF (N)	2026.3	1869.0	108.4%
Net Impulse (N·s)	235.1	253.8	92.6%
Peak RFD (N/s)	9492.0	10472.1	—
Contact Time (s)	0.89	0.84	—
Peak Knee Flexion (°)	54.0	69.1	—
Peak Knee Valgus (°)	2.5	6.8	Δ -4.3°
Peak Tibial IR (°)	-2.3	3.1	Δ -5.4°
Peak Ankle Dorsiflexion (°)	22.0	30.9	Δ -8.8°
Peak Plantarflexor Moment (Nm)	264.5	236.3	111.9%
Peak Hip Adduction (°)	8.2	11.1	Δ -3.0°
Peak Knee Ext Moment (Nm)	161.5	231.0	69.9%
AP COP Excursion (mm)	94.9	154.6	—
Trials analysed	3	3	—
Contact (movement) Time (s)	0.9	0.8	—
Ankle +Work (J)	89.9	161.5	55.7%
Knee +Work (J)	44.1	74.6	59.1%
Hip +Work (J)	301.6	332.2	90.8%
Ankle share of +work (%)	20.6	28.4	—

CLINICAL INTERPRETATION

Isolates each limb — the surgical calf must produce the final push-off without help.

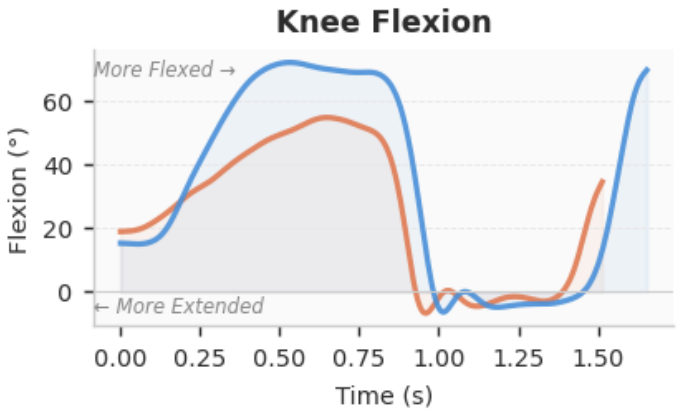
Jump height LSI 86%: ⚠ 14% below non-surgical. Continued plyometric loading required.

Targets: Jump height LSI ≥ 90% · Net impulse LSI ≥ 90% · Heel-raise LSI ≥ 90%.

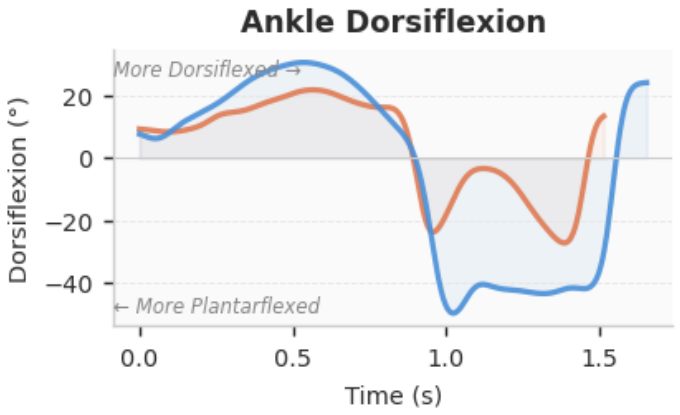


Per-limb GRF (separate trials). A later, drawn-out surgical peak = slow push-off.

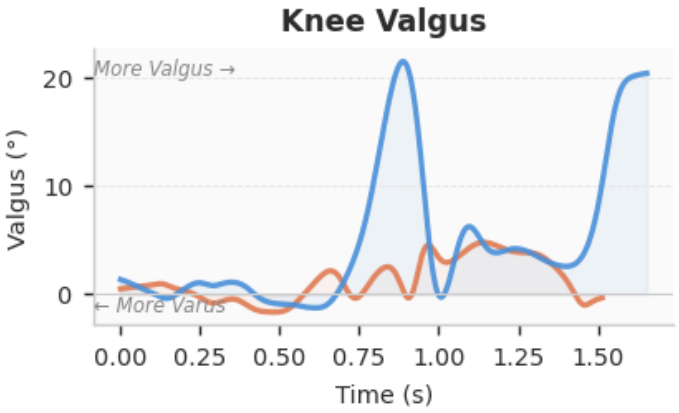
Surgical limb single-leg jump height shows a moderate asymmetry (LSI 86%) — continued loading progression recommended before clearance.



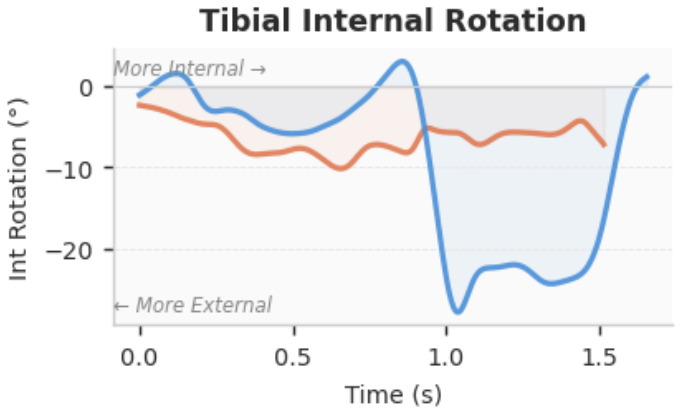
Knee flexion per limb. Reduced surgical depth = shortened, guarded countermovement.



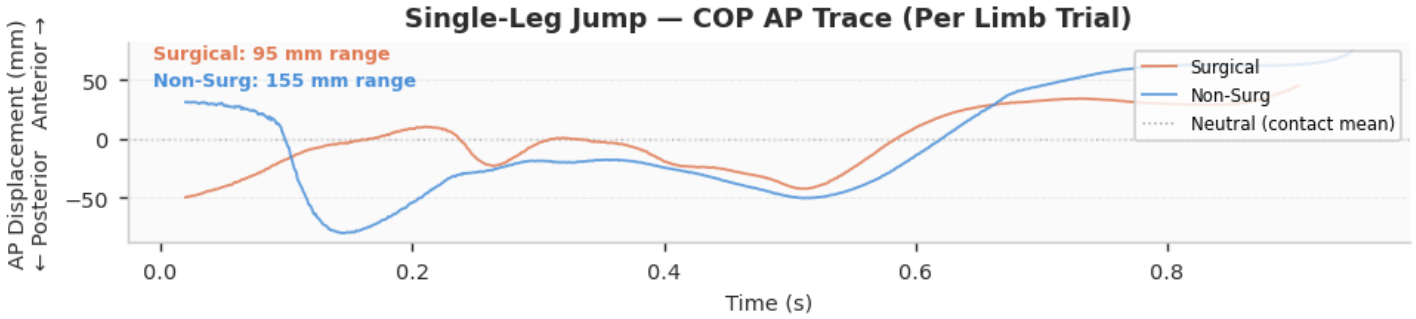
Ankle dorsiflexion per limb — plantarflexion range at push-off drives single-leg height.



Knee valgus per limb (context only for Achilles — not scored).



Tibial / foot rotation per limb. Surgical-side external rotation offset = possible toe-out.



AP COP during the take-off contact (zero-centred). Less anterior travel = limited push-off through the forefoot.

AP COP excursion: surgical 95 mm vs non-surgical 155 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb single-leg jump height shows a moderate asymmetry (LSI 86%) — continued loading progression recommended before clearance.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

ABOUT THIS TEST

Eyes-closed single-leg stance removes visual feedback, so balance depends on ankle and foot proprioception and on the calf's ability to make fast small corrections. Higher COP velocity means more corrective activity. After Achilles repair, reduced calf strength and altered tendon stiffness commonly increase sway on the surgical limb. Limb symmetry (LSI ≥ 85–90%) is the primary criterion.

Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	380.0	140.5	**37.0%**
COP Ellipse Area (mm²)	40703.1	3873.8	**9.5%**
Total Excursion (mm)	2712.6	614.8	—
AP Range (mm)	168.6	106.0	—
ML Range (mm)	366.4	45.0	—
RMS Displacement (mm)	74.7	22.0	—

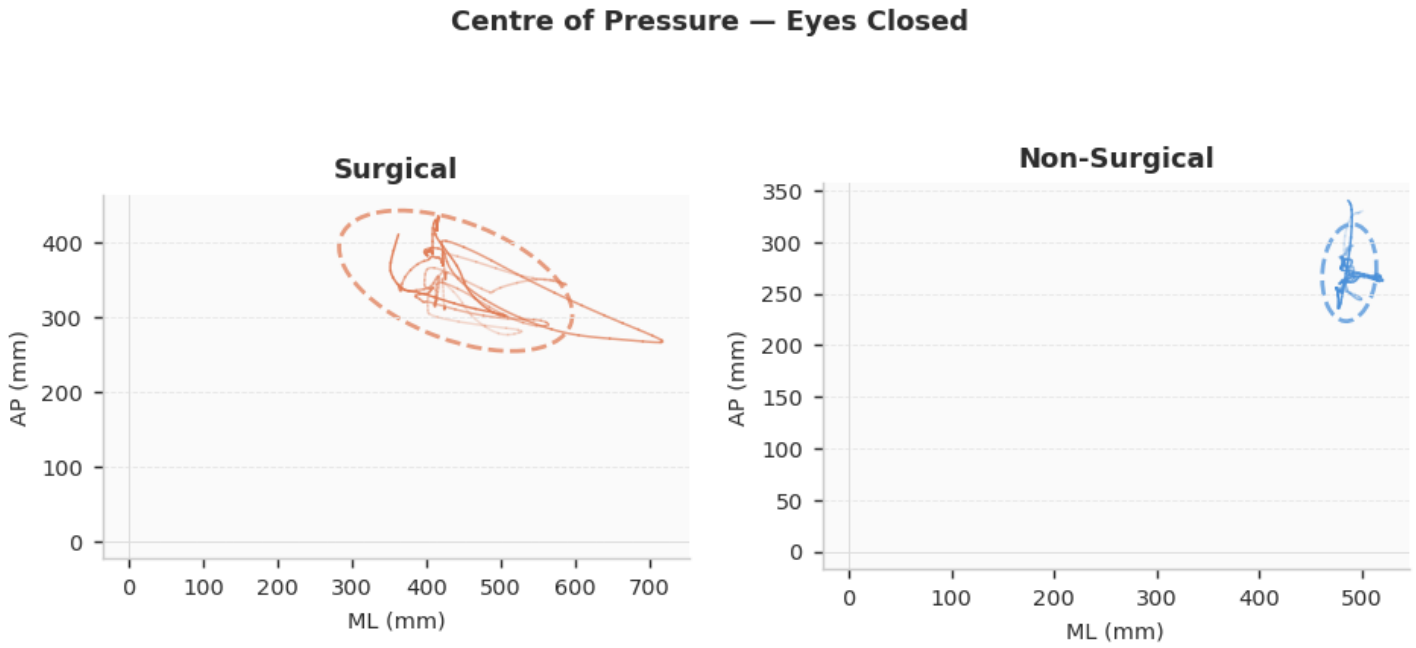
CLINICAL INTERPRETATION

Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception.

Surgical 380 mm/s vs Non-Surgical 140 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 37% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

⚠ **Data check:** COP range on the surgical trial exceeds 150 mm — likely a foot shift or touchdown rather than sway. Review / re-collect before relying on this page.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

ABOUT THIS TEST

Eyes-closed single-leg stance removes visual feedback, so balance depends on ankle and foot proprioception and on the calf's ability to make fast small corrections. Higher COP velocity means more corrective activity. After Achilles repair, reduced calf strength and altered tendon stiffness commonly increase sway on the surgical limb. Limb symmetry (LSI ≥ 85–90%) is the primary criterion.

Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	315.0	225.1	71.5%
COP Ellipse Area (mm²)	33657.9	17728.4	52.7%
Total Excursion (mm)	3634.9	2893.3	—
AP Range (mm)	195.1	191.3	—
ML Range (mm)	221.2	137.2	—
RMS Displacement (mm)	62.8	44.4	—

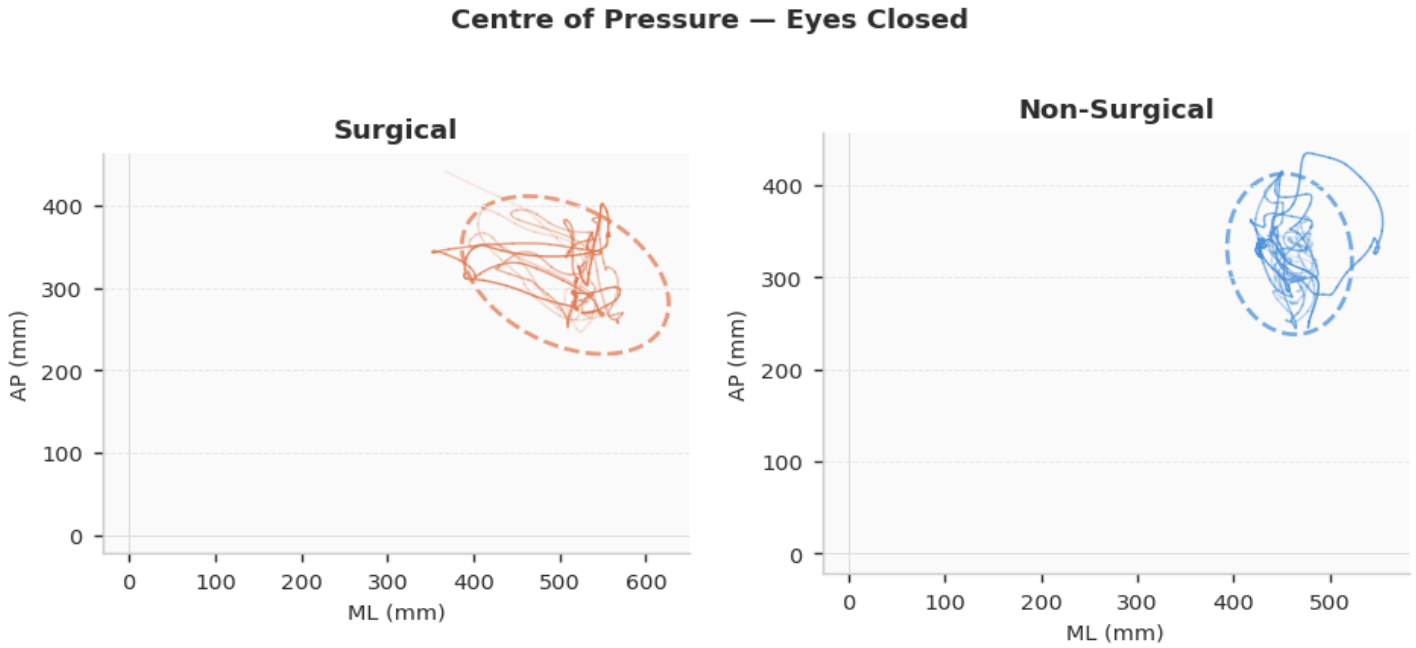
CLINICAL INTERPRETATION

Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 315 mm/s vs Non-Surgical 225 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 71% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

⚠ **Data check:** COP range on the surgical and non-surgical trial exceeds 150 mm — likely a foot shift or touchdown rather than sway. Review / re-collect before relying on this page.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.